

Background

Many people who aim to improve their health, fitness, or dietary habits struggle with accurately tracking their daily intake of micro and macronutrients.

Although nutritional monitoring is essential for achieving goals such as weight management, muscle gain, improved athletic performance, or managing medical conditions, the tools currently used by most individuals are inefficient and unreliable.

Today, tracking nutrients is often done manually or using basic apps that provide limited detail and are not tailored to the user's ever changing nutritional needs. These methods make it easy to miscalculate quantities, skip entries, or lose historical data. As a result, users often fail to notice trends, overlook nutritional deficiencies, or become discouraged by the complexity of the process.

This project aims to solve these issues by creating an intuitive, accurate, and user-friendly platform that enables people to record, analyze, and monitor their nutrient intake over time. Users will be able to log their food quickly, access up-to-date nutrient information, and visualize their progress through historical analytics. This will make nutritional monitoring more accessible, consistent, and effective.

By reducing the friction and errors of traditional methods, the project empowers individuals to make informed decisions about their diet, stay motivated, and reach their goals with confidence.

Goals

1. **Launch Core Features:** The primary objective is to launch the app's core features by the end of February 2026. This includes building the key tools: barcode scanning, manual food entry, and nutrient tracking. This goal will be considered met when the app can successfully calculate and display 100% of macronutrients (protein, carbs, fats) and at least 50% of key micronutrients for each logged food item, relying on the integration of the food facts API. Nutritional data will be obtained through integration with the **Open Food Facts API**, a free and open-source data database that provides detailed macro- and micronutrient information for a wide range of food products. The database is community-driven and continuously expanded thanks to user-contributed data.

2. **Implement the Notification System:** The second objective is to implement a push notification system by mid-March 2026. The purpose is to increase user engagement and help them stay consistent. The goal will be achieved with the integration of two types of functional alerts: a reminder for meal logging and an alert for reaching the daily goal, using standard services for iOS and Android.

3. **Create a Graphical Statistics History:** The third objective is to create a graphical statistics history, by mid-March 2026. This feature will allow users to visualize their progress and habits intuitively, making the data more motivating. This goal will be met when the "Statistics" section displays macronutrient intake through at least three graph types (e.g., bar and line charts) for daily, weekly, monthly and yearly views.

Scope

This project will deliver a mobile app designed to help users monitor their nutrition in a simple and practical way. The app will allow users to scan products' barcodes to instantly view calories and nutrients or manually enter foods when necessary, otherwise, users may add their own recipes, and the ingredients will be added to the calculation of nutritional values based on the quantity of all ingredients contained in the recipe.

The collected data will be organized into daily, weekly, monthly and yearly summaries, supported by clear and intuitive graphs. The collected data will be visualized in a dedicated Statistics section. The Daily view will utilize pie or donut charts to display the percentage distribution of macronutrients (proteins, carbs, fats) against the user's daily goals. Weekly, monthly, and yearly summaries will use bar and line charts to track caloric intake trends and weight progress over time, highlighting days where limits were exceeded. These visual tools will help users to better understand their eating habits and stay aligned with their personal goals, that could be related to weight loss, muscle gain, or general health. A notification will remind users to log their meals.

To complete the project, a preliminary UI/UX design phase is required. This process will start with low-fidelity wireframes and evolve into interactive mockups of the app. The UI will fit the standard smartphone form factor and use standards, it will be easy to interact with, requiring only a few clicks to navigate through. Finally, the app itself will enter development, first, the app's core functionality will be made, and only then, will we integrate access to an open source food facts API. When the app is finalised and testing is complete, the app will be released on both the Google Play Store and the Apple App Store.

The project will not include medical advice, because providing diagnoses or personalized treatments requires qualified medical professionals to avoid the risk of inappropriate care or health complications. Additionally, it will not support languages beyond English and Italian, or advanced features such as food recognition from photos or integration with wearable devices

Key Stakeholders

Mileo Manuela	Commissioner High power / High interest Supportive / Leading
User with specific nutritional needs	End User Low power / High interest Neutral/Supportive
Hospital	End User Low power/High Interest Neutral/Supportive
Project manager	Barakat Ismail (also Backend Manager and Data Engineer)

Exclusions

- **Medical Advice:** We are not providing personalized medical advice or diagnosing health conditions. This exclusion is strictly enforced because clinical guidance falls solely within the competence of licensed medical doctors; providing such advice through an app could lead to inappropriate treatments or jeopardize user health.
- **Languages:** The first version will be in English and Italian only. Multilingual support will be considered only for future updates to minimize initial development complexity.

Project History

Currently the project is in the concept stage. Initial designs and project analysis are underway, but no development work has started yet. The project is starting with the goal of launching a fully functional app by the end of April 2026.

Project Milestones

1. Project Architecture Definition
 - Start: October 21, 2025
 - End: End of November 2025
2. UI/UX Design Approved and Functional Backend Core
 - Start: Early December 2025
 - End: Mid-January 2026
3. Initial Frontend Development
 - Start: Mid-January 2026
 - End: Mid-February 2026
4. Core API Integration (Goal 1 complete)
 - Start: Mid-February 2026
 - End: End of February 2026
5. Advanced Feature Development (Goals 2 & 3 complete)
 - Start: Early March 2026
 - End: Mid-March 2026
6. Beta Testing and Feedback Collection
 - Start: Mid-March 2026
 - End: Early April 2026
7. Final Bug Fixing and Release
 - Start: Early April 2026
 - End: End of April 2026

Project Budget

Estimated Budget: €124

Resources include:

- Play Store License: €25 (one-time fee)
- Apple Store License: €99 / year
- OpenFoodFacts API (a free Food Data API, less robust than paid alternatives): €0
- PlanetScale (User Database, "hobby" plan: 5GB storage and 1 billion reads/month): €0
- Vercel or Netlify (Backend API Hosting, Free Plan): €0

Equipment: Development will be carried out using the personal equipment (laptops and mobile test devices) belonging to the project team members. No new equipment purchases are budgeted.

Spaces: The team will work remotely from personal workspaces and utilize school facilities for meetings and collaboration. No budget is allocated for dedicated office space.

People: The core project team is defined in the Key Stakeholders section. No budget is allocated for "extra people," such as external consultants or contractors.

Funding will be provided by the project sponsor and institutional partners.

Constraints, Assumptions, Risks and Dependencies

Constraints	• Dependency on third-party APIs: The project relies on the OpenFoodFacts API, which may have limitations on data accuracy or request rates. • Time constraints: The project is linked to the school calendar • App Store Policies: The app must adhere strictly to the review guidelines and policies of both the Apple App Store and the Google Play Store.
Assumptions	• User Hardware: Users will have access to modern smartphones (iOS/Android) capable of running the app. • Free-Tier Sufficiency: It is assumed that the free-tier plans for PlanetScale (database) and Vercel/Netlify (API hosting) will remain available and will be sufficient to handle the app's initial user load. • API Functionality: Third-party APIs (OpenFoodFacts) and SDKs (barcode scanning) will remain functional and accessible. • User Base: The user base will initially focus on Italy and English-speaking regions.
Dependencies	• External Service Availability: The project's core functionality depends on the continued operation of PlanetScale, Vercel/Netlify, and the OpenFoodFacts API. • Third-Party SDKs: Relies on third-party SDK support for barcode scanning within Flutter. • App Store Approval: The final release is entirely dependent on a successful review and approval process by Apple and Google.
Risks	• Service Changes or Failures: Key services (OpenFoodFacts, PlanetScale, Vercel) could change their terms, introduce breaking API changes, or experience downtime. • Exceeding Free Tier: A major risk is that user adoption is too successful, causing the app to exceed the free-tier limits of PlanetScale or Vercel, which would create unexpected costs. • App Store Rejection: Risk of app rejection or significant delays during the

	Apple/Google store review process, which could impact the final deadline.
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- Team Availability: Delays in development due to limited team availability.
- User Engagement: Low user engagement if the UI is not intuitive enough.

Approval Signatures

[Name], Project Client

[Name], Project Sponsor

Barakat Ismail
[Name], Project Manager